

The nice thing about cloud chamber tracks is we can tell just by their appearance what class they fall into. These are cosmic ray tracks, so for the most part they will either be electron or muon tracks.

$\frac{dE}{dx}$ represents the rate at which an ionizing particle loses energy as it travels through a material. This directly corresponds to the track's brightness and width. Minimum-ionizing particles (muons, fast electrons) leave faint, thin tracks. Slow or highly charged particles (alphas, low-energy protons) leave thick, dense tracks.

The RMS scattering angle per unit path length scales as $1/(p \cdot \beta)$, and for a given momentum, a lighter particle scatters much more than a heavier one at the same momentum, mainly because low-momentum electrons dominate the secondary population. In practice: muons stay straight, electrons wander.

Muon Tracks

Feature	Typical Appearance
Width	Thin, uniform, minimum-ionizing
Straightness	Long, straight
Length	Long, often spans the full visible chamber depth
Density	Uniform along the whole path
Branching	None, or very rarely

Electron Tracks

Feature	Typical Appearance
Width	Thin, but often fainter/patchier than a muon track
Straightness	kinks, zig-zags, curls
Length	Shorter and more variable; can be short stubby fragments
Density	Uneven, broken-looking, denser near abrupt direction changes
Branching	Can show delta-ray-like forks or a "starburst" of short segments radiating from one point

Other: Proton or Alphas (RARE)

Feature	Typical Appearance
Width	Thick, dense. Noticeably heavier than muon or electron lines
Straightness	Straight, short
Length	Stubby
Endpoints	Often show a distinct thickening at the terminus